MBEYA UNIVERSITY OF SCIENCE AND TECHNOLOGY

COLLEGE OF INFORMATION AND COMMUNICATION TECHNOLOGY

DEPARTMENT OF ELECTRONICS AND TELECOMMUNICATION ENGEERING

DEVELOPING AN ARTIFICIAL INTELLIGENCE (AI) MODEL FOR DETECTING TUBERCULOSIS FROM CHEST X-RAY IMAGES.

Student Name	: RAPHAEL E. RAPHAEL
Registration Number	:22100934340085
UE Number	: UE/BETS/25/14537
Course Name	: PROJECT I
Course Code	:ET 8409
Supervisor	: Dr CHARLES NYATENGA
Academic Year	:2025/2026

March, 2026

A Project Report Submitted in Partial Fulfillment of the Requirements for Bachelor of Engineering in Telecommunication Systems of Mbeya University of Science and Technology

DEVELOPING AN ARTIFICIAL INTELLIGENCE (AI) MODEL FOR DETECTING TUBERCULOSIS FROM CHEST X-RAY IMAGES.

By

RAPHAEL E. RAPHAEL

22100934340085

Mbeya University of Science and Technology

College of Information and Communication Technology

Department of Electronics and Telecommunication Engineering

P.O Box 131, Mbeya

March, 2026

CERTIFICATION

The undersigned hereby certify that the project titled" **Developing an Artificial Intelligence (Ai) Model for Detecting Tuberculosis from Chest X-Ray Images."** submitted in partial fulfillment of the requirements for the Bachelor of Engineering in Telecommunication Systems at Mbeya University of Science and Technology, has been reviewed and is recommended for acceptance.

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Name of Supervisor:

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Signature:

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Date:

LIST OF ABBREVIATIONS

AI: Artificial Intelligence

TB: Tuberculosis

CNN: Convolutional Neural Network

CXR: Chest X-Ray

DL: Deep Learning

ML: Machine Learning

CAD: Computer-Aided Diagnosis

ViT: Vision Transformer

GUI: Graphical User Interface

IoT: Internet of Things

XAI: Explainable Artificial Intelligence

CT: Computed Tomography

SVM: Support Vector Machine

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CHAPTER ONE
INTRODUCTION

1.1 Background of the Study

Tuberculosis (TB) is a serious infectious disease that mainly affects the lungs. It is caused by a bacterium called Mycobacterium tuberculosis. The disease spreads through the air when an infected person coughs, sneezes, or talks.

TB remains one of the major health problems in many developing countries, especially in Africa and Asia. Millions of people are infected every year, and many deaths occur due to late diagnosis and delayed treatment.

Early detection of tuberculosis is very important because it helps doctors start treatment quickly and prevent the disease from spreading to other people. One of the most common diagnostic tools used to detect TB is chest X-ray imaging. Chest X-ray images allow doctors to observe abnormalities in lung tissues that may indicate TB infection.

However, the interpretation of chest X-ray images requires trained specialists known as radiologists. In many hospitals, especially in rural areas of developing countries, there is a shortage of radiologists. This shortage leads to delays in diagnosis and treatment.

Manual analysis of X-ray images also takes a lot of time and may sometimes lead to human errors. Radiologists must carefully analyze each image to detect abnormalities, which can be difficult when there are many patients waiting for diagnosis.

Recent advancements in Artificial Intelligence (AI) and deep learning have made it possible to develop automated systems that can analyze medical images. Deep learning models, especially Convolutional Neural Networks (CNNs), have shown excellent performance in image classification and medical diagnosis.

CNN models can automatically learn important features from images and identify patterns associated with diseases. These models have been successfully applied in detecting lung diseases from chest X-ray images.

Therefore, developing an AI model that can analyze chest X-ray images and detect tuberculosis can help improve diagnostic accuracy, reduce workload for radiologists, and support healthcare systems in developing countries.

1.2 Problem Statement

Tuberculosis continues to be a major public health challenge worldwide. Early diagnosis is important for reducing the spread of the disease and improving patient survival rates (WHO, 2023). Despite the availability of chest X-ray imaging technology, many hospitals face challenges in diagnosing TB effectively. One of the major problems is the shortage of trained radiologists who can analyze chest X-ray images. This shortage is more serious in rural hospitals and developing countries (Wang, X., 2022)

Another challenge is that manual interpretation of X-ray images is time-consuming. Radiologists must examine each image carefully, which can delay diagnosis when the number of patients is large (Jaeger et al., 2021).

In addition, human errors may occur during the diagnosis process. Small abnormalities in chest X-ray images may sometimes be missed, which may lead to incorrect diagnosis and delayed treatment.

Due to these challenges, there is a need for an automated system that can assist doctors in detecting tuberculosis quickly and accurately. Artificial Intelligence and deep learning technologies provide an opportunity to develop such systems.

1.3 Main Objective

The main objective is: To develop an Artificial Intelligence (AI) model for detecting tuberculosis from chest X-ray images.

1.4 Specific Objectives

To achieve the main objective, the study will focus on the following specific objectives:

1. To collect and prepare a chest X-ray dataset

This involves gathering chest X-ray images from reliable medical datasets. The images must be organized and labeled correctly as TB positive or TB negative. Data preparation may also involve cleaning the dataset, resizing images, and improving image quality for training the AI model.

2. To design and train a deep learning model

In this step, a Convolutional Neural Network (CNN) model will be designed. The model will be trained using the prepared dataset so that it can learn patterns associated with TB infection in chest X-ray images.

3. To evaluate the performance of the model

After training, the model will be tested using unseen data. Performance metrics such as accuracy, precision, recall, and F1-score will be used to measure how well the model can detect TB.

1.5 Significance of the Study

This study is significant because it addresses several important challenges in tuberculosis diagnosis. Firstly, the developed system can help doctors detect TB faster by automatically analyzing chest X-ray images. Manual examination of X-rays is time-consuming and requires highly trained radiologists, but an AI system can quickly process images and provide preliminary results, saving valuable time in busy healthcare settings. Secondly, the system can support hospitals with a shortage of radiologists. In many developing countries, including Tanzania, some hospitals and rural clinics do not have enough specialized staff to handle large volumes of X-ray examinations. The AI model can assist healthcare workers in such environments, providing a second opinion or preliminary analysis.

Thirdly, the system can reduce the time required for diagnosis, allowing patients to receive treatment earlier. Early detection of TB is critical because it prevents the disease from worsening and reduces the risk of spreading it to others. Fourthly, the model may improve diagnostic accuracy and reduce human errors. Even experienced radiologists may miss subtle signs of TB in X-rays, especially when images are of low quality. By using a consistent AI-based approach, diagnostic errors can be minimized. Finally, the developed model has the potential to be especially useful in rural hospitals and healthcare centers where medical resources are limited. By providing an affordable and accessible tool for TB detection, the system can contribute to better healthcare delivery and disease management in underserved areas.

1.6 Scope of the Study

This study focuses on the development of an AI model for detecting tuberculosis from chest X-ray images. The main tasks involved in this project include collecting chest X-ray datasets, which will provide the data necessary to train the AI model, and training a deep learning model, specifically a convolutional neural network (CNN), to recognize features associated with TB. Once the model is trained, it will be tested to evaluate its performance, ensuring that it can accurately classify Xrays as TB-positive or TB-negative. In addition, the project will involve developing a simple prediction interface, which will allow healthcare staff to easily input X-ray images and receive diagnostic predictions from the system.

The study is limited to TB detection and does not aim to diagnose other lung diseases such as pneumonia, lung cancer, or COVID-19. This focused scope ensures that the AI model is optimized specifically for TB detection, improving its accuracy and reliability. While the system can be extended in the future to include other lung conditions, the current study prioritizes developing a lightweight, efficient, and practical tool for TB diagnosis that can be used in resource-limited healthcare settings.

1.7 Organization of the Study

This report is structured into five chapters to provide a clear and logical presentation of the research. Chapter One introduces the study, including the background, problem statement, research objectives, significance, and scope. This chapter sets the context and explains why the study is necessary. Chapter Two presents the literature review, discussing previous research on TB detection using AI and deep learning models. It highlights the strengths, weaknesses, and gaps in existing studies, providing a foundation for the current research.

Chapter Three describes the research methodology, including system design, dataset preparation, model development, and testing procedures. This chapter explains how the AI model is built and evaluated. Chapter Four presents the results of the developed system, analyzing its performance and discussing the findings in detail. Finally, Chapter Five provides the conclusion of the study and offers recommendations for future work, including possible improvements to the AI model and its deployment in healthcare settings. This organization ensures that the report follows a logical flow from introduction and background to development, results, and final conclusions.

CHAPTER TWO
LITERATURE REVIEW

2.1 Introduction

This chapter reviews previous studies related to tuberculosis detection using artificial intelligence and deep learning techniques. The purpose of the literature review is to examine the existing research conducted in the field of medical image analysis and to identify the methods used in detecting tuberculosis from chest X-ray images.

The chapter also discusses the technologies used in previous studies, including deep learning algorithms such as Convolutional Neural Networks (CNNs). By reviewing these studies, it becomes possible to identify their strengths, limitations, and research gaps that motivate the development of the proposed system.

Artificial intelligence techniques have increasingly been applied in healthcare to improve disease diagnosis and assist doctors in analyzing medical images (Oloko-Oba & Viriri, 2022)

2.2 Theoretical Review

2.2.1 Tuberculosis and Chest X-Ray Diagnosis

Tuberculosis is an infectious disease caused by Mycobacterium tuberculosis that primarily affects the lungs. The disease spreads through airborne particles when an infected person coughs or sneezes. Chest X-ray imaging is one of the most widely used tools for TB screening because it allows doctors to observe lung abnormalities such as cavities, nodules, or infiltrates. These patterns help doctors determine whether a patient may have tuberculosis.

Recent studies have shown that chest X-ray images can be effectively used to train artificial intelligence models for automated TB detection (Williams et al., 2023)**.**

However, interpreting chest X-ray images requires trained radiologists, and many hospitals experience a shortage of these specialists. As a result, automated diagnostic systems using artificial intelligence have been proposed to support medical professionals.

2.2.2 Artificial Intelligence in Medical Imaging

Artificial Intelligence (AI) refers to computer systems that can perform tasks that normally require human intelligence, such as learning, pattern recognition, and decision making.

In medical imaging, AI techniques are used to analyze images such as X-rays, CT scans, and MRI scans to detect diseases. Machine learning and deep learning methods allow computers to learn from large datasets of medical images and identify patterns associated with specific diseases.

Recent research shows that AI models can significantly improve disease detection accuracy and reduce diagnostic time in medical imaging systems (Wang, X., 2022)AI-based systems can assist doctors by providing preliminary diagnoses, highlighting abnormal regions in medical images, and improving overall healthcare efficiency.

2.2.3 Deep Learning in Medical Image Analysis

Deep learning is a branch of machine learning that uses neural networks with multiple layers to analyze complex data. Deep learning models have achieved significant success in image classification and object detection tasks. In healthcare applications, these models can analyze medical images to detect diseases such as cancer, pneumonia, and tuberculosis.

One of the advantages of deep learning models is their ability to automatically extract important features from images without manual feature engineering.

Recent studies show that deep learning techniques have achieved high accuracy in detecting tuberculosis from chest X-ray images (Showkatian et al., 2022)

2.2.4 Convolutional Neural Networks (CNN)

Convolutional Neural Networks are a type of deep learning model specifically designed for image analysis. CNN models consist of several layers that process images step by step. These layers include convolution layers, pooling layers, and fully connected layers.

The convolution layer extracts important features such as edges, shapes, and textures from the image. The pooling layer reduces the size of the feature maps while preserving important information.

Finally, the fully connected layer performs classification and produces the prediction result. CNN models have been widely used for medical image analysis because they can achieve high performance in detecting diseases from X-ray images.

Studies have demonstrated that CNN-based systems can successfully detect tuberculosis patterns in chest radiographs with high accuracy (Mirugwe et al., 2025)

2.3 Conceptual Framework

A conceptual framework describes the structure of a research study and shows the relationship between the main components of the system. It explains how the input data is processed to produce the final output of the system.

In this study, the conceptual framework explains how chest X-ray images are processed using an Artificial Intelligence model to detect tuberculosis.

The framework consists of three main components: Input, Processing, and Output.

2.3.1. Input (Chest X-ray Images)

The input of the system is chest X-ray images collected from medical datasets. These images represent the lung condition of patients and may include both normal lungs and lungs infected with tuberculosis. Before the images are used in the AI model, they must undergo data preprocessing. Preprocessing may include resizing the images, removing noise, normalizing pixel values, and organizing the dataset into training and testing groups.

Proper data preparation is important because it improves the quality of the dataset and helps the AI model learn more effectively.

2.3.2. Processing (Deep Learning Model)

The processing stage is the core part of the system. In this stage, the chest X-ray images are analyzed using a deep learning model, specifically a Convolutional Neural Network (CNN). CNN models are widely used for image recognition tasks because they can automatically extract important features from images such as edges, textures, and shapes (LeCun et al., 2020). The CNN processes the images through several layers: Convolution layers, which detect important image patterns. Pooling layers, which reduce image size while preserving important features. Fully connected layers, which perform classification based on the extracted features. During the training process, the model learns patterns that are associated with tuberculosis infection. After training, the model can analyze new chest X-ray images and classify them accordingly.

Deep learning methods have shown strong performance in medical image analysis and can help improve diagnostic accuracy.

2.3.3. Output (Prediction Results)

The output of the system is the classification result produced by the AI model. After analyzing the chest X-ray image, the system predicts whether the patient is: TB Positive, meaning the model detects signs of tuberculosis infection. TB Negative, meaning the lungs appear normal without TB infection. The prediction results can assist doctors in making faster diagnostic decisions and can also serve as a supportive tool in hospitals with limited radiology experts.

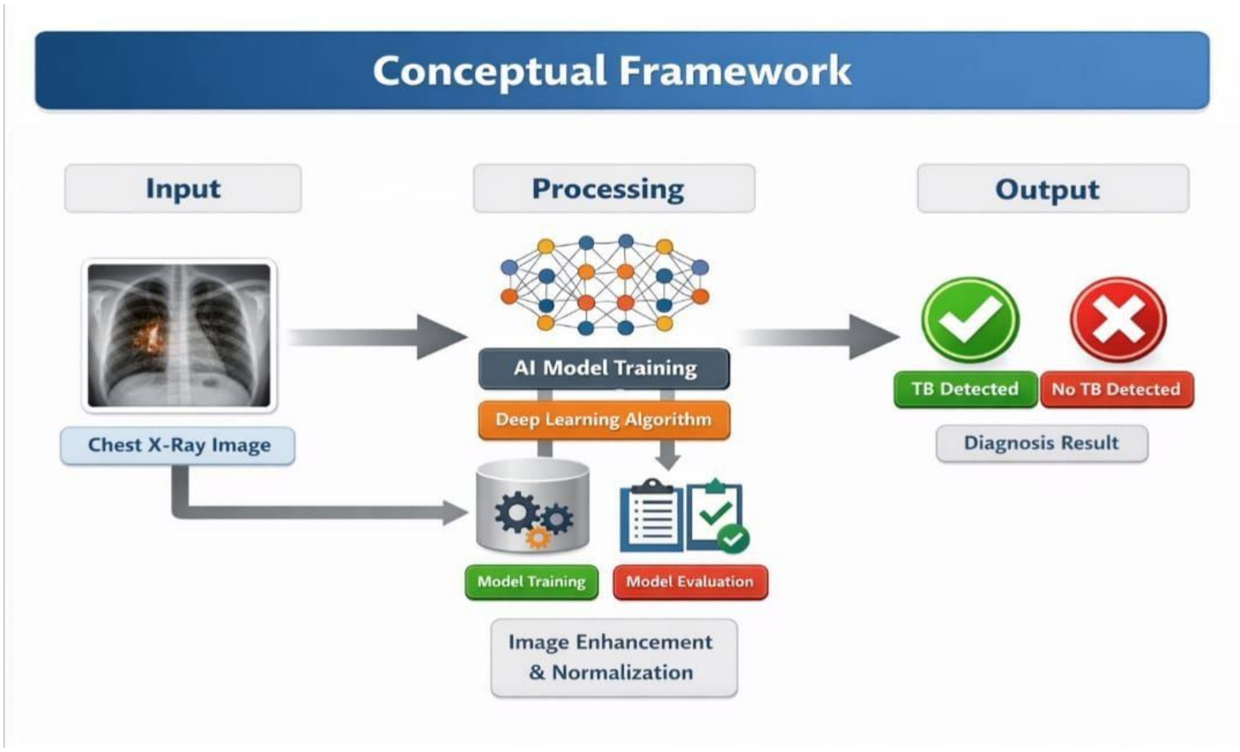

Figure 1 conceptual framework:

2.4 Review of Related Works

Several researchers have explored the use of artificial intelligence (AI) for detecting tuberculosis (TB) from chest X-ray images, highlighting both the potential and the limitations of current systems.

Kassim, Y., Abdullah, N., & Ahmad, (2022) developed a TB detection system using ensemble deep learning models, which combine multiple neural networks to improve prediction accuracy. Ensemble models are effective because they leverage the strengths of different models, reducing errors caused by individual networks. However, the system required high computational power and advanced hardware such as GPUs and large memory capacity. This dependence on expensive infrastructure makes it difficult to deploy the system in small clinics or rural hospitals, which often have limited computing resources.

Arora et al., (2023) used EfficientNet-B7, a state-of-the-art deep learning architecture designed for high performance in image classification tasks. While the model achieved excellent results on high-quality and clean datasets, its performance decreased when handling noisy images or those captured under poor lighting conditions. Such conditions are common in real hospital environments, indicating that the model may lack robustness for practical deployment. (Huang et al., 2021) applied the Vision Transformer (ViT) model to TB detection. Vision Transformers are particularly powerful because they can capture global relationships and patterns in images, unlike conventional convolutional neural networks (CNNs) that focus primarily on local features. Despite its high accuracy, the ViT model requires very large datasets and powerful GPUs for training, making it expensive and difficult to deploy in resource-limited settings.

(Abbasi et al., 2025)proposed a hybrid CNN-Transformer model that combines the local feature extraction strengths of CNNs with the global pattern recognition capabilities of Transformers. This hybrid approach improved detection performance compared to single-model architectures. However, the system lacked optimization for TB-specific features and still required complex computational resources, limiting its usability in clinics with constrained infrastructure. Overall, these studies show that while AI can enhance TB detection, there is a consistent trade-off between model complexity, computational requirements, and real-world applicability, particularly in developing countries.

2.5 Empirical Review

Empirical studies are important because they show how different researchers have developed and tested systems for tuberculosis (TB) detection using artificial intelligence (AI). These studies provide real experimental results such as accuracy, usability, and system performance. By reviewing empirical studies, researchers can understand what has already been achieved and what problems still remain. This section discusses previous research based on four major aspects: effectiveness of compared systems, usability and accessibility, challenges and barriers, and recommendations for future research.

(i) Effectiveness of Compared Systems

Several studies have investigated the effectiveness of AI models for detecting tuberculosis from chest X-ray images. Most of these systems rely on deep learning techniques, especially Convolutional Neural Networks (CNNs), because they are very good at analyzing medical images.

A study conducted by (Kassim, Y., Abdullah, N., & Ahmad, 2022)developed a TB detection system using ensemble deep learning models. Ensemble learning combines several neural networks in order to improve prediction accuracy. The researchers trained their system using a large dataset of chest X-ray images and evaluated the performance using accuracy and sensitivity metrics. The system achieved high detection accuracy above 90%, showing that combining multiple models can improve the reliability of TB diagnosis.

Similarly(Arora et al., 2023)) proposed a system based on the EfficientNet-B7 deep learning architecture for detecting TB from chest X-ray images. Efficient Net is known for its ability to achieve high accuracy with optimized computational efficiency. Their experimental results showed that the model achieved strong classification performance and was able to correctly identify TB cases in many test images. This demonstrates that modern deep learning architectures can significantly improve automated disease detection.

Another study by (Lu et al., 2024)applied Vision Transformer (ViT) models to TB detection. Transformers are commonly used in natural language processing but have recently been applied to image analysis. Their results showed that the transformer-based approach achieved competitive accuracy compared to CNN-based models. The study demonstrated that attention-based models can capture complex image patterns that may be missed by traditional convolutional networks.

In addition(Shahzad et al., 2025)introduced a hybrid CNN-Transformer model that combines convolutional feature extraction with transformer attention mechanisms. The results indicated that the hybrid model improved classification performance and achieved better generalization across different datasets.

Overall, empirical evidence shows that AI-based systems are highly effective in detecting tuberculosis from chest X-ray images. Many systems achieve accuracy levels above 85–95%, which indicates strong potential for assisting medical professionals in TB screening (Bappi et al., 2023)However, despite these promising results, some studies note that the effectiveness of AI models depends on factors such as dataset quality, model complexity, and computational resources.

(ii) Usability and Accessibility of the Systems

While many AI systems show strong performance in research experiments, usability and accessibility are also important factors, especially in developing countries where medical resources may be limited.

Usability refers to how easy a system is for healthcare professionals to operate. Some TB detection systems require advanced computing knowledge to operate the AI models, which may limit their practical use in hospitals. For example, Bappi et al., (2023)reported that many AI diagnostic systems are designed mainly for research environments rather than real clinical settings. This makes them difficult for non-technical users such as nurses or general practitioners.

Accessibility refers to how easily healthcare facilities can obtain and use the technology. In many developing countries, including regions in Africa, hospitals may not have access to high performance computing equipment**.** Qin et al., (2021) emphasized that although deep learning models can achieve high accuracy, they often require powerful GPUs and large memory capacity. Some researchers have attempted to improve accessibility by developing lightweight AI models that can run on normal computers or mobile devices, (Shariful et al., 2023) proposed a simplified CNN model that reduces computational complexity while maintaining acceptable diagnostic accuracy. Such systems are more suitable for small clinics and rural healthcare centers.

Furthermore, integrating AI systems with simple graphical user interfaces (GUIs) can improve usability. A user-friendly interface allows healthcare workers to upload chest X-ray images and receive predictions without needing specialized technical training.

Therefore, although many TB detection systems are effective, improving usability and accessibility remains essential to ensure that these technologies can be applied in real healthcare environments.

(iii) Challenges and Barriers

Despite the progress in AI-based TB detection, several challenges and barriers still exist. One major challenge is the availability of high-quality datasets. Deep learning models require large datasets for training. However, medical image datasets are often limited due to privacy concerns and ethical restrictions. According to **Pasa et al. (2020),** small or imbalanced datasets can lead to biased models that may not perform well in real-world conditions.

Another challenge is high computational requirements. Advanced models such as Vision Transformers and deep CNN architectures require powerful GPUs and large amounts of memory for training and inference. This makes them difficult to implement in hospitals with limited technological infrastructure **(Zhang & Li, 2024).**

A further barrier is lack of generalization across different populations and imaging conditions. Many AI models are trained using datasets from specific countries or hospitals. When applied to other environments, the system may produce lower accuracy because of differences in X-ray machines, image quality, or patient demographics (Arora et al., 2023)

Additionally, there are ethical and regulatory challenges associated with AI in healthcare. Medical institutions must ensure that AI systems are reliable, safe, and compliant with healthcare regulations before deployment. Trust from healthcare professionals is also necessary for successful adoption of AI technologies.

Finally, limited technical expertise in some healthcare institutions can slow the implementation of AI systems. Without trained personnel to manage and maintain the systems, hospitals may struggle to adopt AI-based diagnostic tools.

(iv) Recommendations for Future Research Directions

Based on the limitations observed in previous studies, several recommendations have been proposed for future research.

First, researchers should focus on developing lightweight and efficient AI models that can run on low-cost computers. Such models will improve accessibility in small clinics and rural hospitals where advanced computing infrastructure is not available.

Second, future studies should prioritize training models using diverse datasets from different geographic regions. This will improve the generalization ability of AI systems and make them more reliable for global healthcare use.

Third, integrating explainable AI (XAI) techniques is important. Explainable AI helps doctors understand how the model makes predictions, which can increase trust and transparency in AIassisted diagnosis (Samek et al., 2021)**.**

Another important direction is the development of mobile and cloud-based TB detection systems. These systems allow healthcare workers to upload X-ray images using mobile devices and receive results from cloud-based AI models. Such technology could significantly improve TB screening in remote areas.

2.6 Existing System:

Convolutional Neural Network (CNN) Based TB Detection System

One of the most common existing systems used for detecting tuberculosis (TB) from chest X-ray images is the Convolutional Neural Network (CNN) based detection system. A CNN is a type of deep learning model that is widely used in medical image analysis because it can automatically learn patterns from images.

In this system, chest X-ray images are first collected from hospitals or medical databases. These images are then preprocessed to improve their quality by resizing them, normalizing the pixel values, and removing noise. After preprocessing, the images are fed into the CNN model for training. The CNN contains several layers such as convolution layers, pooling layers, and fully connected layers. These layers work together to extract important features from the X-ray images, such as lung shapes, shadows, and abnormal patterns that may indicate tuberculosis infection. During the training process, the model learns to distinguish between normal lungs and TB-infected lungs by analyzing thousands of labeled X-ray images. After training, the system can analyze new chest X-ray images and automatically classify them as TB positive or TB negative. Many researchers have reported that CNN-based TB detection systems can achieve high accuracy and can support doctors in making faster diagnoses (Pasa et al., 2019)**.**

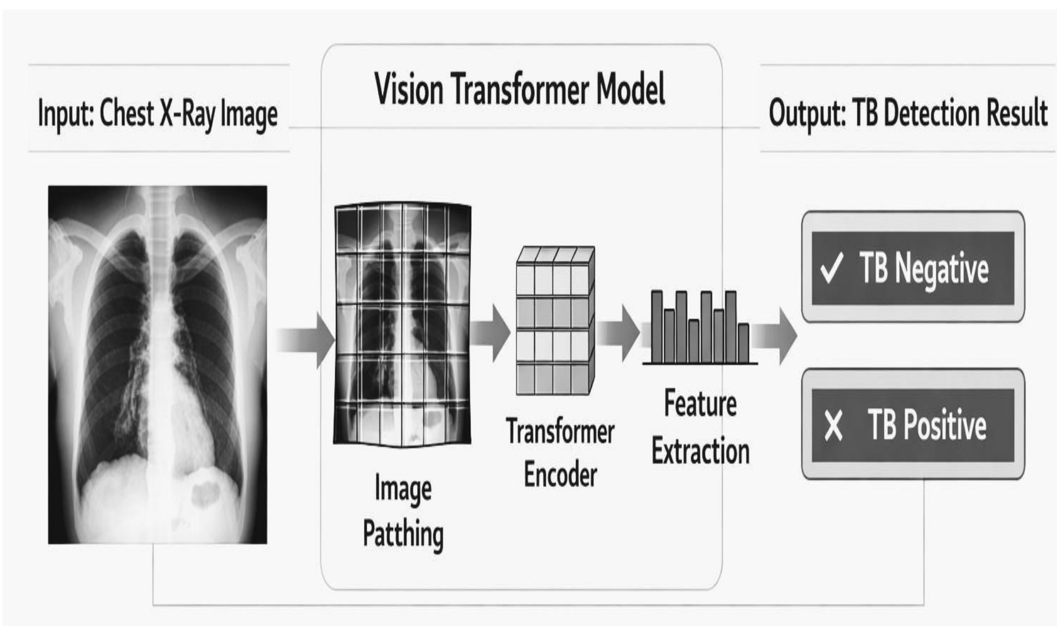

Figure 2: exiting model

2.6.1 Drawbacks of exiting system

(i) High Computational Requirements

One major drawback of CNN-based TB detection systems is that they require high computational power for training and sometimes for prediction. Deep learning models contain many parameters that must be learned during training. This process requires powerful hardware such as Graphics Processing Units (GPUs) and large memory capacity.

In many hospitals, especially in developing countries, such advanced computing resources are not available. Small clinics may only have basic computers that cannot efficiently run deep learning models. As a result, implementing CNN-based TB detection systems in these environments becomes difficult. This limitation reduces the accessibility of the technology for healthcare facilities with limited resources.

(ii) Drawback 2: Dependence on Large and High-Quality Datasets

Another important limitation of CNN-based systems is that they require large and high-quality datasets to achieve good performance. Deep learning models learn patterns from data, and if the dataset is small or poorly labeled, the model may not learn the correct features for detecting TB. In medical imaging, collecting large datasets is often difficult due to privacy issues, ethical restrictions, and limited availability of labeled data. When a CNN model is trained with insufficient data, it may produce inaccurate predictions or fail to generalize well when used in real clinical environments. This can reduce the reliability of the system when diagnosing patients from different hospitals or regions.

Summary

In summary, CNN-based TB detection systems are widely used because they can automatically analyze chest X-ray images and achieve high diagnostic accuracy. However, the system has some limitations, including high computational requirements and dependence on large datasets. These challenges highlight the need for developing simpler and more efficient AI models that can operate effectively in healthcare environments with limited resources.

2.7 Research Gap

Although many researchers have developed artificial intelligence (AI) systems for detecting tuberculosis (TB) from chest X-ray images, several limitations still exist in the current systems. Identifying these limitations helps to understand what has not yet been fully addressed and why further research is needed. This section explains the research gap that motivates the development of the proposed system.

First; many existing TB detection systems are designed using very complex deep learning architectures, such as ensemble models, large convolutional neural networks, and transformer based models. While these systems can achieve high accuracy, they often require high computational power and expensive hardware such as GPUs to train and run the models. In many developing countries, including Tanzania, hospitals and clinics may not have access to such advanced computing resources. Because of this limitation, these AI systems cannot be easily implemented in local healthcare environments where resources are limited.

Second; many of the existing models rely on very large and clean datasets to achieve good performance. However, in real hospital settings, chest X-ray images may have different qualities due to variations in imaging equipment, patient movement, or environmental

Another limitation is that some existing systems are not designed with user-friendly interfaces. Many research models focus mainly on improving accuracy but do not consider how healthcare workers will interact with the system. Doctors, nurses, or technicians in hospitals may not have advanced technical knowledge to operate complex AI systems. Without simple interfaces, these systems become difficult to use in daily medical practice.

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